

UNIT 03 [Advanced]

↳ Chapter 01 : Electricity.

LESSON 01

Control the Magnet.

Time Frame: [90 mins]

Lesson Objectives:

1. Understand the principles of electromagnetism and its application in cranes.
2. Create a simple electromagnet using basic materials.
3. Integrate a joystick and servo motor to control crane movement.

Challenge Questions:

1. How can you make your electromagnet stronger?
2. What happens when you disconnect the battery, and why?
3. How can you improve the crane's design to lift heavier loads?

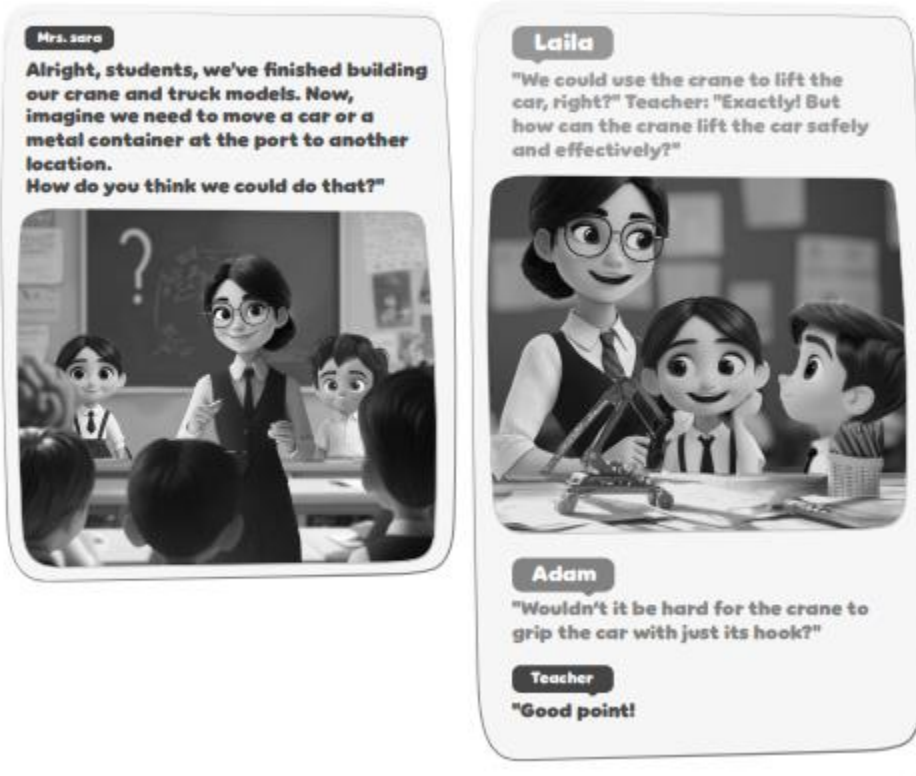
SEL Relation:

1. Encourages teamwork and problem-solving in hands-on projects.
2. Builds resilience through trial and error in experimentation.
3. Enhances communication skills through discussions and project presentations.

Lesson Plan: Electricity & Magnetism in Crane Design

1. Engage

- **Teacher Role:**
 - Guide a discussion on how cranes pick up metal containers.
 - Ask thought-provoking questions to spark curiosity about electromagnets.
- **Expected From Students:**
 - Participate in discussions, share ideas, and ask questions.



- **Intro Story:**
 - Continue the story of Laila and Adam designing a crane.
 - Discuss how cranes use magnets to grab metal containers.
- **Story Reflection:**
 - **Questions:**
 1. How do cranes lift and drop metal objects?
 2. What materials are affected by magnets?
 - **Suggested Answers:**
 - Cranes use electromagnets that turn on and off.
 - Magnets attract iron and other metals.
 - **Strategy:** Think-Pair-Share: Students think individually, discuss in pairs, and share with the class.

2. Think



let's Think

How can the crane lift the car safely and effectively?

Do you think we can turn magnets on and off?

- **Questions and Possible Answers:**
 - What is an electromagnet? (A magnet that is activated by electricity.)
 - Why do some materials stick to a magnet while others don't? (Only certain metals are magnetic.)
- **Brainstorm Web:**
 - Students list ideas on how magnets can be used in real life.
 - Group discussion on different ways to power a magnet.
- **Strategy:** KWL Chart (What we Know, Want to know, Learned).

3. Explore

- **Teacher Role:**
 - Guide hands-on experimentation with electromagnets.
- **Expected From Students:**
 - Follow instructions, observe, and discuss results.

Explore 



Let's create our own magnet, wrap a copper wire around a nail and connect it to a battery.

And approach it to these things.

Write your observation:



- **Activity - Do It:**

- **Materials Needed:** Copper wire, nail, battery, paper clips.
- **Step-by-Step Instructions:**
 1. Wrap the copper wire around the nail.
 2. Connect the wire to the battery terminals.
 3. Test if the nail picks up paper clips.
 4. Disconnect the battery and observe if the magnetism disappears.
- **Reflection Questions:**
 - What happens when the battery is disconnected?
 - How could we use this in a crane design?

4. Explain

- **Teacher Role:**
 - Explain the science behind electromagnets.
 - Guide discussion on how electromagnets are used in real life.
- **Expected From Students:**
 - Watch, listen, and discuss.



- **Watch It:**
 - **Pre-Watching:** Ask students what they expect to learn.
 - **During Watching:** Take notes on how electromagnets work.
 - **Post-Watching:** Discuss key takeaways.
 - **Video Link:** <https://www.youtube.com/watch?v=tHlchO1pbFA>
- **Read About It:**
 - **Pre-Reading:** Discuss real-world applications of electromagnets.
 - **During Reading:** Identify key concepts.
 - **Post-Reading:** Summarize the main points.

5. Elaborate

- **Teacher Role:**
 - Encourage students to apply their knowledge creatively.
- **Expected From Students:**
 - Design and build an improved electromagnet crane.



Coin



Eraser



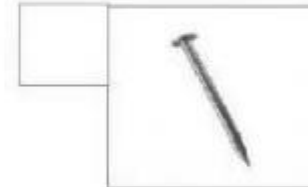
Lollipop



**Paper
clip**



**Plastic
spoon**



Nail

Disconnect the wire from the battery and approach these things again.

Write your observation:

- **Assessment:**
 - **Focus Questions:**
 - How can we control when the magnet turns on and off?
 - What happens if we use a stronger battery?
 - **Practice:**
 - **Materials Needed:** Joy stick, servo motor.
 - **Step-by-Step Instructions:**
 1. Connect a joystick to control crane movement.
 2. Attach a servo motor for rotation.
 3. Test and refine the design.

Practice



Create an electromagnet and add it to your crane.



Joy stick can be added to the crane to control it and make it move right and left.



Servo motor is motor that move 180°.



After adding electromagnet to your crane, go back and write that in the improve part.



- **Reflection Questions:**

- What challenges did you face?
- How did you improve your design?

- **Challenge:** Improve the crane design using the Engineering Design Process (EDP) worksheet.

6. Evaluate

- **Teacher Role:**

- Assess projects and provide constructive feedback.

- **Expected From Students:**

- Demonstrate understanding by explaining and showcasing their projects.

Assessment

Focus



Choose the correct answer from the choices for each question:

1. What is an electromagnet?
 - A) A magnet made from plastic.
 - B) A magnet created using electricity.
 - C) A type of metal.
 - D) A kind of food.

2. Which of the following materials is best for making an electromagnet?
 - A) Iron.
 - B) Wood.
 - C) Plastic.
 - D) Glass.

3. Can an electromagnet be turned on and off?
 - A) No, it is always on.
 - B) Yes, by connecting and disconnecting the battery.
 - C) No, it is always off.
 - D) Yes, by changing its color.

- **Showcase:**
 - Students present their electromagnet-powered cranes.
 - Evaluate using a rubric based on functionality, creativity, and explanation.

Relation to Standards:

- Understanding electromagnetism in practical applications.
- Applying engineering principles to design functional models.
- Developing critical thinking through experimentation and problem-solving.